

Algebra 1
Unit 4
Lesson 12 Practice

Name Answer Key
Date _____
Period _____

1. Write the equation of the line that passes through (3,2) and is parallel to

$$3x - 2y = 1.$$

$$\begin{array}{r} -3x \quad -3x \\ -2y = \frac{1}{-2} - \frac{3x}{-2} \\ y = \frac{3}{2}x - \frac{1}{2} \end{array}$$

$$m = \frac{3}{2} \quad \begin{matrix} x & y \\ (3, 2) \end{matrix}$$

$$y - 2 = \frac{3}{2}(x - 3) \quad \text{OR}$$

$$y = \frac{3}{2}x - \frac{5}{2}$$

$$\begin{array}{r} y - 2 = \frac{3}{2}x - \frac{9}{2} \\ +2 \quad +2 \end{array}$$

Use the line shown on the coordinate grid to complete questions 2 through 4.

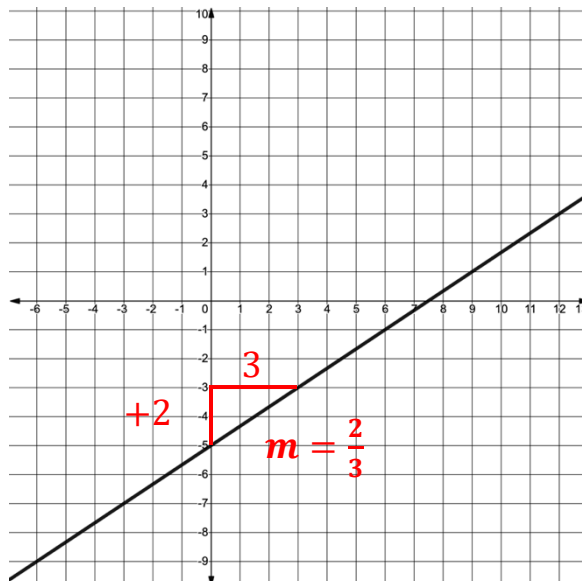
2. What is the slope of the line that is perpendicular to the line shown on the coordinate grid?

$$m = -\frac{3}{2}$$

3. Write the equation of the line that is perpendicular to the line shown on the coordinate grid and that passes through (1,0).

$$\begin{matrix} x & y \\ y - 0 = \frac{-3}{2}(x - 1) \end{matrix}$$

$$y = -\frac{3}{2}x + \frac{3}{2}$$



4. Write the equation of the line that is parallel to the line shown on the coordinate grid and that passes through (5, -4)

$$m = \frac{2}{3}$$

$$y + 4 = \frac{2}{3}(x - 5) \quad \text{or} \quad y = \frac{2}{3}x - \frac{22}{3}$$

5. Write the equation of the line in standard form that is perpendicular to $y = -\frac{1}{4}x - 13$ and passes through (1,7).

$$\perp m = 4$$

$$y - 7 = 4(x - 1)$$

$$y - 7 = 4x - 4$$

$$\begin{array}{r} -4x \quad -4x \\ y - 4x - 7 = -4 \end{array}$$

$$y - 4x - 7 = -4$$

$$+7 \quad +7$$

$$-4x + y = 3 \quad \text{or} \quad 4x - y = -3$$

Use the following information about Line U to complete questions 6 through 8.

Line U passes through $(-2, -2.5)$ and $(1, 0)$.

6. Line Z is parallel to Line U . What is the slope of Line Z ?

$$\begin{array}{c|c} x & y \\ \hline -2 & -2.5 \\ 1 & 0 \end{array}$$

$+3 \swarrow \quad \searrow +2.5$

$$\frac{\Delta y}{\Delta x} = \frac{2.5}{3} = \frac{5}{6}$$

$$\text{Slope of line } Z = \frac{5}{6}$$

7. Line Z passes through $(-6, 9)$. Write the equation of Line Z in slope-intercept form.

$$y - 9 = \frac{5}{6}(x + 6)$$

$$\begin{array}{rcl} y - 9 & = & \frac{5}{6}x + 5 \\ +9 & & +9 \end{array}$$

$$y = \frac{5}{6}x + 14$$

8. Line C passes through $(7, 3)$ and is perpendicular to Line U . Write the equation of Line C in point-slope and standard forms. $\perp m = -\frac{6}{5}$

Point-Slope Form	Standard Form
$y - 3 = -\frac{6}{5}(x - 7)$	$\begin{aligned} (5)y - 3(5) &= \frac{-6(5)}{5}x + \frac{42(5)}{5} \\ 5y - 15 &= -6x + 42 \\ +6x + 15 &+ 6x + 15 \\ \hline 6x + 5y &= 57 \end{aligned}$

9. Write the equation of the line that is parallel to $-21x + 18y = 3$ and passes through point $(-6, -3)$ in point-slope form.

$$m = \frac{7}{6}$$

$$y + 3 = \frac{7}{6}(x + 6)$$

$$\begin{array}{rcl} +21x & & +21x \\ 18y & = & 3 + 21x \\ \hline 18 & = & \frac{3}{18} + \frac{21x}{18} \\ & & \frac{7}{6}x + \frac{1}{6} \\ y & = & \frac{7}{6}x + \frac{1}{6} \end{array}$$

10. Write the equation of the line that is perpendicular to $\frac{10x}{-1} = \frac{-y}{-1}$ and passes through point $(4, \frac{2}{5})$.

$$\perp m = \frac{1}{10}$$

$$y - \frac{2}{5} = \frac{1}{10}(x - 4)$$

$$y = -10x$$